

ENV 6106

Air Pollution Modeling

Spring 2023

Topics

- AERMET overview
- Expectation
 - Understand the role of AERMET and its components
 - Know the inputs requirement for AERMET and data types
 - Understand the three stages of AERMET processing
 - Understand important pathways in AERMET inputs
- Readings: AERMET User guides
 - Available at <https://www.epa.gov/scram>

AERMOD

- American Meteorological Society/Environmental Protection Agency Regulatory Model
- EPA recommended/preferred dispersion model for near field applications
 - Replaced Industrial Source Complex (ISC) in 2006
 - ISC has a life-span of > 20 years, very widely used
- Refined model applications
 - Recall AERSCREEN is the screening version

Components of AERMOD

- AERMET: meteorological data preprocessor
 - AERSURFACE: for surface characteristics
 - AERMINUTE: for 1-min ASOS data
- AERMAP: terrain data preprocessor
- AERMOD: uses input data (meteorological, terrain data etc) to perform refined analysis
 - BPIPPRIM: a multi-building dimensions program incorporating the GEP technical procedures for building downwash applications.

AERMET overview

AERMET

- Process surface and upper air data for use by AERMOD
- Requires:
 - Surface characteristics: albedo etc
 - Surface meteorological observations
 - Hourly National Weather Services (NWS) observations
 - Automated Surface Observation System (ASOS) data
 - Upper air soundings (Radiosondes, Rawinsondes)
 - Usually 00:00 and 12:00 GMT; ± 1 hour common
 - Special soundings may available for other times
 - Or, site-specific data for surface
 - Data collected by a facility, may not be on-site, no fixed format, need to let AERMET know what variables are in there
 - Or, data from numerical weather model (such as WRF)
 - Require other tools; also commercially available

AERMET

- The major function of AERMET is to
 - Extract and QA raw, archived observational data
 - Upper air, surface, and on-site data
 - Calculate required meteorological parameters for AERMOD modeling needs
 - 1-min ASOS data processed by AERMINUTE were combined here (already QA'ed)
 - Output to two files: surface and profile

Past 3-stage processing

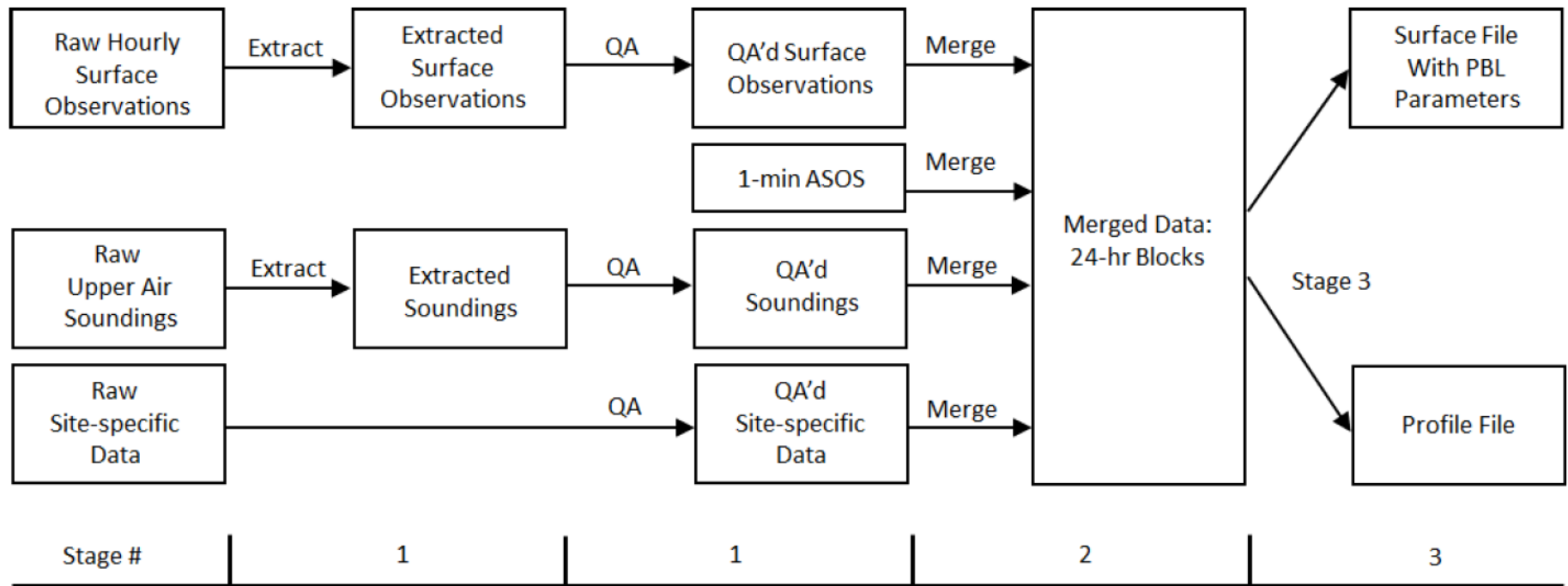


Figure 1-1. AERMET Processing

Now 2-stage processing

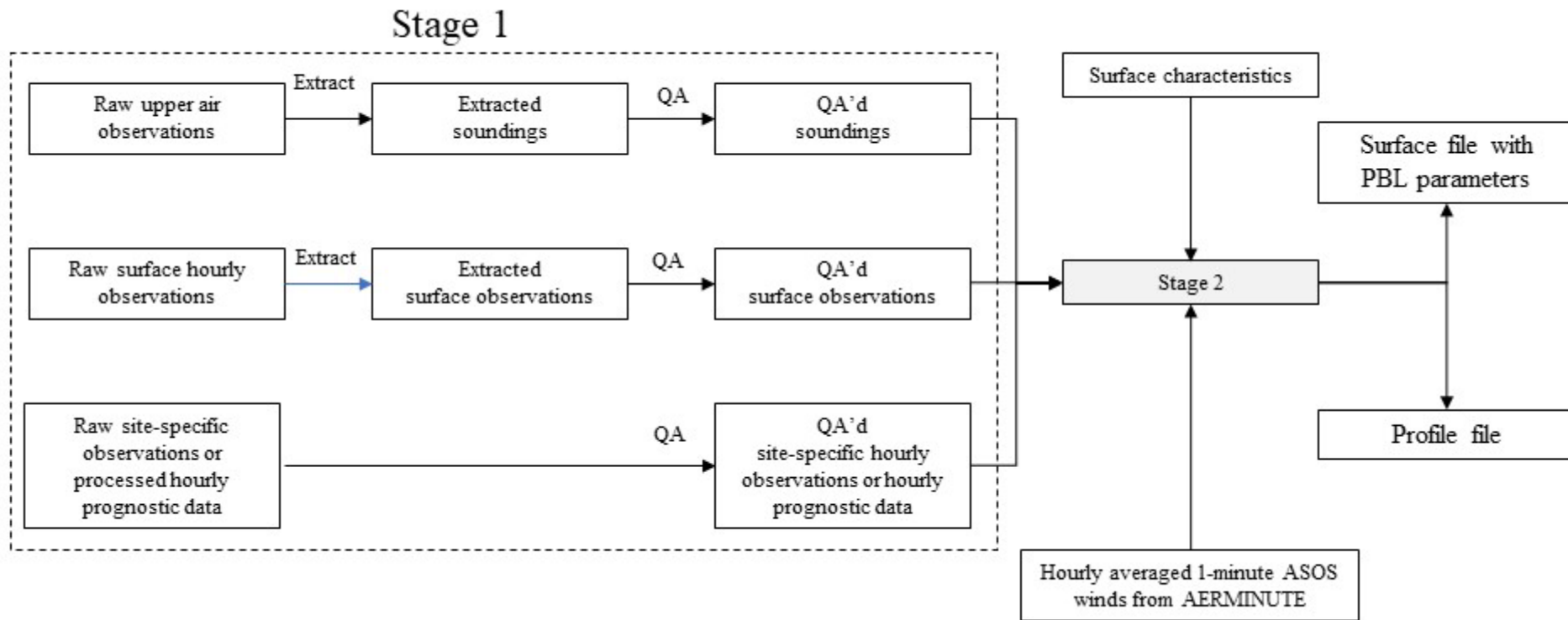


Fig 1-1 AERMET user guide

2-stage processing

- Two control files for 2 stages
- Each control file contains a few pathways
 - Pathways defines the nature of task
- Each pathway contains a few keywords
 - Keywords defines details of a task
- Analogy:
 - “Pathway”: Prepare-for-work
 - “Keywords”: wake up, clean up, dress up, commute to work

Input Data



A surface ASOS station

By Famartin (Own work) [CC BY-SA 3.0 (<https://creativecommons.org/licenses/by-sa/3.0>)], via Wikimedia Commons

What is AWOS: <https://www.youtube.com/watch?v=JrIA RiUoZs8>

Call 407-855-5235 for Orland International Airport ASOS auto voice service

Gone with the wind – Wind Rose

Horizontal mean flow

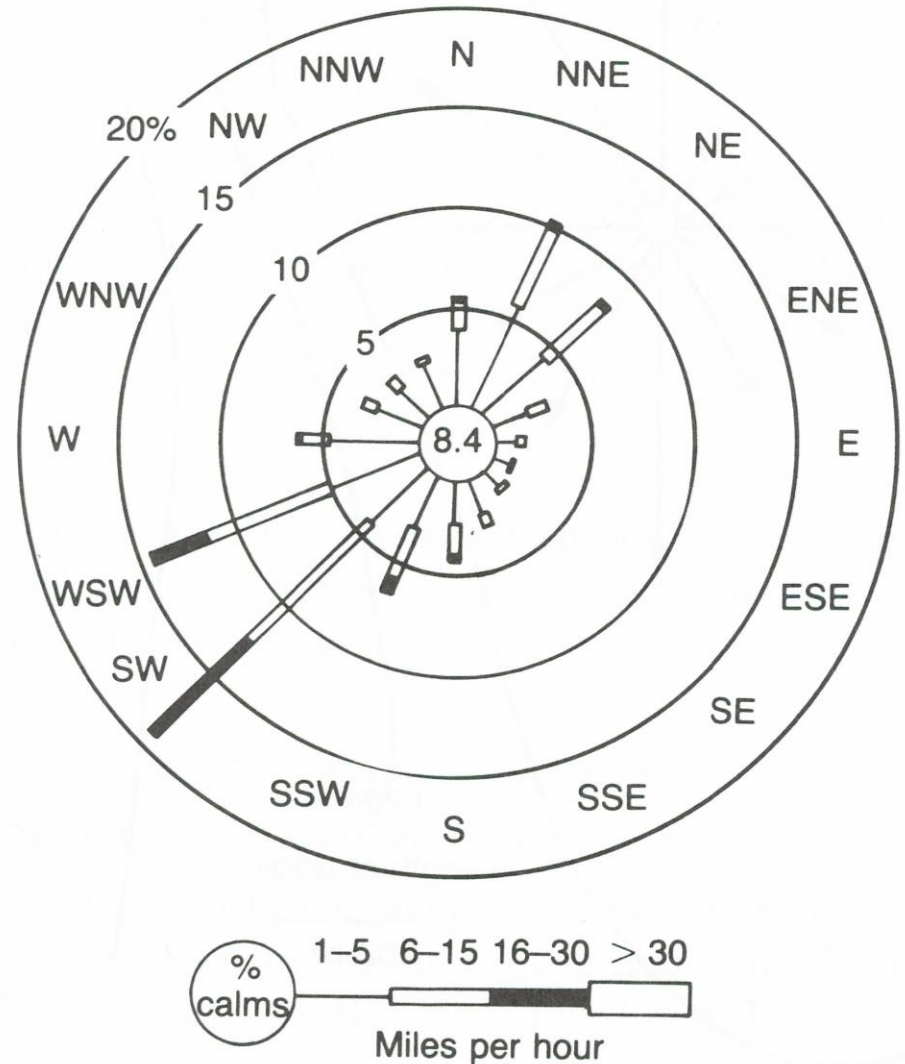
Direction

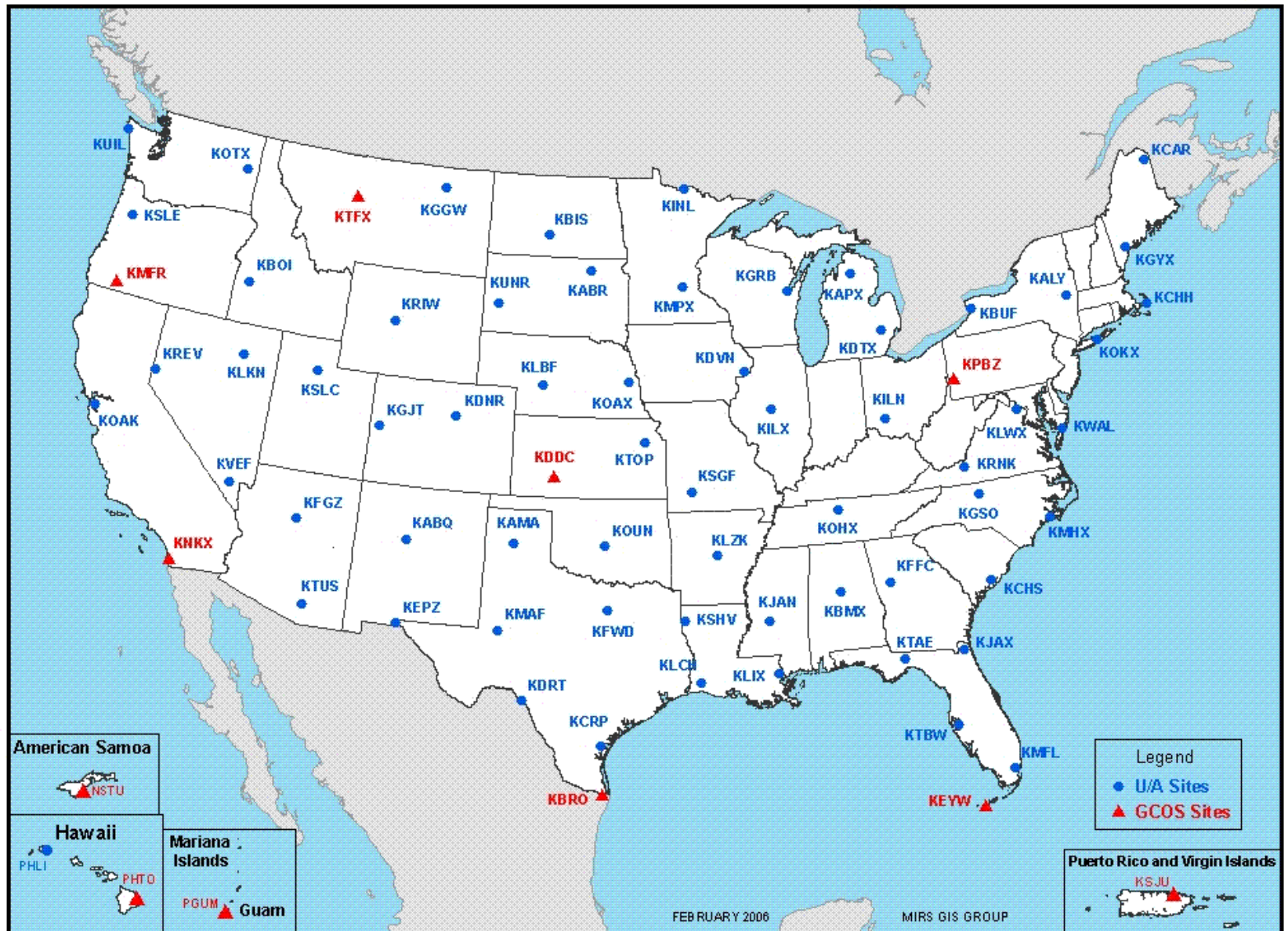
Where wind is coming from
(angle in degrees clockwise
from the north)

Direction of pollutant transport

Speed

transports pollutants further
dilutes emissions
creates turbulent mixing
bends plumes





NWS Rawinsonde Network (https://www.weather.gov/upperair/nws_upper)

Contents of surface data

- Pressure
- Temperatures
- Precipitation (type and intensity)
- Humidity
- Wind direction, speed and character (gusts etc)
- Cloud height and quantity
- Visibility, including fog, haze, dust
- More information maybe collected such as start and end time of rain etc

Format of surface data

- ISHD (ISH, ISD) product. Integrated Surface Hourly Data
 - Compilation of global hourly surface observations from > 35,000 stations worldwide
 - TD-3505 format. Reports in GMT time
 - Do not use abbreviated ISHD
- ASOS 1-minutes wind data
 - Automated sensor collected data. > 900 stations in US
 - TD-6405 format.
 - Has to be pre-processed using AERMINUTE

Format of surface data

- AERMET also accepts other, some are really old, formats
 - CD-144: ended 12/31/95
 - SCRAM (1/1/1984 – 12/31/1992)
 - SAMSON (1/1/1961 – 12/31/1990)
 - HUSWO (1/1/1990 – 12/31/1995)
 - TD-3280
- Some of these data only available on CD/DVD and even on tapes

Format of upper air data

- Collected from twice a day soundings
 - Pressure, temperature etc at many heights
 - Report by pressure levels
 - Mandatory and Significant Pressure Levels
 - Up to ~100mb level (~16 km). 1mb = 100Pa
 - Do include all levels in modeling (Mand & Sigs)
 - 1990-present, past data available by order
 - Integrated Global Radiosonde Archive (IGRA) from NCEI not supported

Where are the data

- One met stations may has several IDs
 - Station call sign (e.g. KMCO)
 - US Air Force ID (e.g. 722050)
 - WBAN (Weather-Bureau-Army-Navy) ID (e.g. 12815)
 - WMO (World Meteorological Organization) ID
- Surface, ISHD hourly data
 - TD-3505 format
 - Most up to date full ISHD inventory:
<ftp://ftp.ncdc.noaa.gov/pub/data/noaa/isd-history.csv>
 - Explore data at: <https://www.ncei.noaa.gov/access/search/data-search/global-hourly>
 - Find a station: use “Map View”
 - Name format: USAF ID + “-” + WBAN ID + “-” + Year
 - Check if format is correct!

Where are the data

- Surface, 1-min ASOS data
 - TD-6405 format
 - <ftp://ftp.ncdc.noaa.gov/pub/data/asos-onemin/>
 - Look for folders start with 6405
 - File format: 6405 + 0 + station call sign + YYYY + MM
 - To find a station near your study area:
 - https://www.faa.gov/air_traffic/weather/asos/
 - Look for only ASOS stations
- Upper air sounding data
 - TD-6201 is outdated! Use FSL format
 - Download from: <https://ruc.noaa.gov/raobs/>
 - 1990-present, past data available by order

AERMET

- Requires:
 - Surface characteristics
 - Surface observations
 - Hourly National Weather Services (NWS) observations
 - Automated Surface Observation System (ASOS) data
 - Upper air soundings (Radiosondes, Rawinsondes)
 - Usually 00:00 and 12:00 GMT; ± 1 hour common
 - Special soundings may available for other times
 - Or, site-specific data for surface
 - Data collected by a facility, may not be on-site, no fixed format, need to let AERMET know what variables are in there
- FDEP pre-processed AERMOD ready data for Florida:
 - <https://floridadep.gov/air/air-business-planning/content/aermet-datasets-map>

Components of AERMOD

- AERMET: meteorological data preprocessor
 - AERSURFACE: for surface characteristics
 - AERMINUTE: for 1-min ASOS data
- AERMAP: terrain data preprocessor
- AERMOD: uses input data (meteorological, terrain data etc) to perform refined analysis
 - BPIPPRIM: a multi-building dimensions program incorporating the GEP technical procedures for building downwash applications.

AERMET Stage 1

Now 2-stage processing

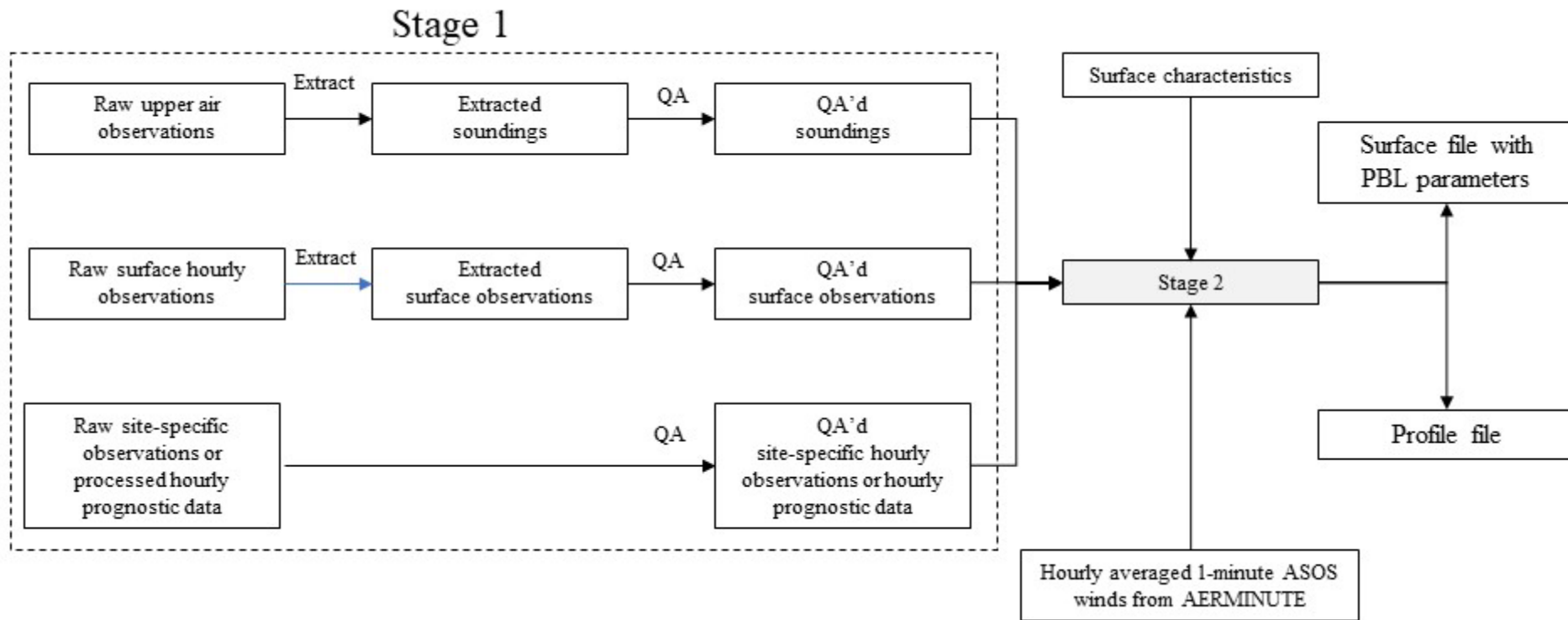


Fig 1-1 AERMET user guide

Stage 1

- Extract and QA raw, archived observations
 - Surface, upper air, and on-site data
- Allowable pathways
 - JOB: messages and reports
 - SURFACE: process surface data
 - UPPERAIR: process upper air data
 - ONSITE: process on-site data

Stage 1 Pathway: JOB

- Most useful Keywords:
 - MESSAGES: log file
 - Format: MESSAGES *log_file_name*
 - REPORT: optional, summary report
 - Format: REPORT *summary_file_name*
- Example

JOB

MESSAGES	TestRun.MSG
REPORT	TestRun.RPT

Stage 1 Pathway: SURFACE

- Most useful Keywords:
 - DATA: Archived surface data name and format
 - Format: DATA *file_name file_format*
 - EXTRACT: Name of file contains **all** extracted data
 - Format: EXTRACT *extracted_file_name*
 - QAOUT: Name of file contained **QA'ed** data
 - Format: QAOUT *qa_data_filename*
 - You can choose not to use either all data record, or only QA'ed data for later calculation

Stage 1 Pathway: SURFACE

- Most useful Keywords:
 - LOCATION: coordinates of surface station
 - Format: LOCATION *site_id lat lon* [timeadjust] [elevation]
 - Lat and lon can be reverse, need to be positive numbers
 - Timeadjust and elevation (meters) are optional
 - ISHD is reported in GMT time, all others in local time
 - Thus timeadjust is set to 0 for data other than ISHD
 - Timeadjust is **reverse of standard** timezone #
 - Specify **5** for eastern US (UTC-5), not daylight time
 - XDATES: Range of date to extract
 - Format: XDATES *YB/MB/DB YE/ME/DE*
 - 4 digits year, 2 or 1 digits month and day
 - 2 digits year no longer works

Stage 1 Pathway: SURFACE

- Example

SURFACE

DATA	722050-12815-2022 ISHD
EXTRACT	MCO_2022_Extract.dat
QAOUT	MCO_2022_QAout.dat
LOCATION	722050 28.434N 81.325W 5 0
XDATES	2022/1/1 2022/12/31

Avoid using “tab” in input file

Stage 1 Pathway: UPPERAIR

- Most useful Keywords:
 - DATA: Archived upper air data name and format
 - Format: DATA *file_name file_format*
 - EXTRACT: Name of file contains **all** extracted data
 - Format: EXTRACT *extracted_file_name*
 - QAOUT: Name of file contained **QA'ed** data
 - Format: QAOUT *qa_data_filename*

Stage 1 Pathway: UPPERAIR

- Most useful Keywords:
 - LOCATION: coordinates of upper air station
 - Format: LOCATION *site_id lat lon timeadjust*
 - Lat and lon can be reverse, need to be positive numbers
 - Timeadjust is **not** optional
 - All upper air data reported in GMT time
 - Timeadjust is **reverse of standard** timezone #
 - Specify **5** for eastern US (UTC-5), not daylight time
 - XDATES: Range of date to extract
 - Format: XDATES *YB/MB/DB YE/ME/DE*
 - 4 digits year, 2 or 1 digits month and day
 - 2 digits year no longer works

Stage 1 Pathway: UPPERAIR

- Example

UPPERAIR

DATA	TPA_UpperAir.txt FSL
EXTRACT	UA_2022_Extract_FSL_knot.dat
QAOUT	UA_2022_QAout_FSL_knot.dat
LOCATION	12868 28.48N 80.55W 5
XDATES	2022/1/1 2022/12/31

Avoid using “tab” in input file

Stage 1 Pathway: ONSITE

- ONSITE pathway tells AERMET how to read on-site meteorological observations
- This pathway is a bit complicated because data collected from different locations have different formats
- We'll not go through specific keywords in this class
- More details see section 4.5 of AERMOD guide

Stage 1 example input

JOB

MESSAGES	TestRun.MSG
REPORT	TestRun.RPT

SURFACE

DATA	722050-12815-2022 ISHD
EXTRACT	MCO_2022_Extract.dat
QAOUT	MCO_2022_QAout.dat
LOCATION	722050 28.434N 81.325W 5 0
XDATES	2022/1/1 2022/12/31

UPPERAIR

DATA	TPA_UpperAir.txt FSL
EXTRACT	UA_2022_Extract_FSL_knot.dat
QAOUT	UA_2022_QAout_FSL_knot.dat
LOCATION	12868 28.48N 80.55W 5
XDATES	2022/1/1 2022/12/31

AERMET Stage 2

Now 2-stage processing

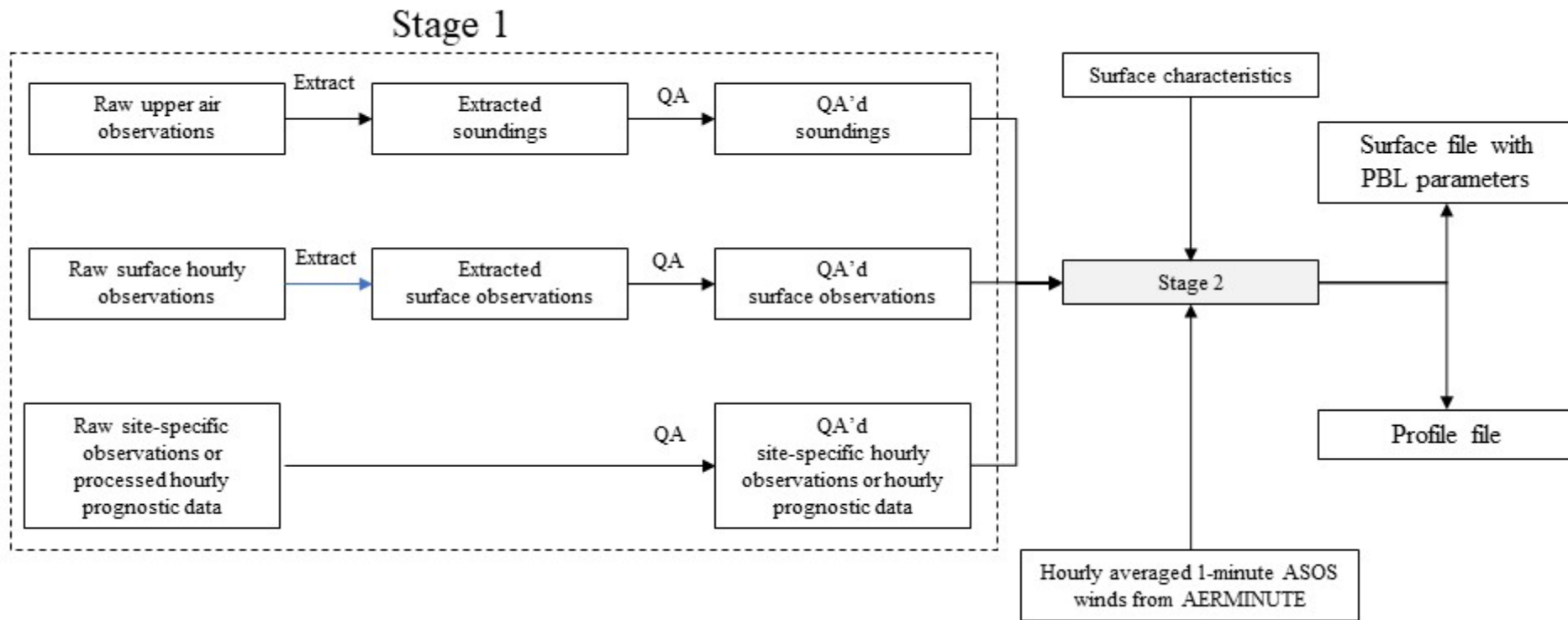


Fig 1-1 AERMET user guide

Stage 2

- Using stage 1 data, calculate meteorological parameters for AERMOD
- Combined past stage 2 & 3
- Allowable pathways
 - JOB: messages and reports
 - SURFACE: specific stage 1 surface data
 - UPPERAIR: specific stage 1 upper air data
 - METPREP: prepare met data

Stage 2

- JOB pathway
 - MESSAGES: log file
 - REPORT: optional, summary report
- SURFACE pathway
 - QAOUT: QA'ed, or not QA'ed, surface data
 - ASOS1MIN: AERMINUTE processed ASOS data
- UPPERAIR pathway
 - QAOUT: Similar with SURFACE

Stage 3 pathway: METPREP

- METPREP pathway
 - DATA: stage 2 merged file
 - XDATES: time range of processing
 - OUTPUT: output file for surface data
 - PROFILE: output file for profile (upper air) data
 - Specify surface characteristics
 - **FREQ_SECT**, **SECTOR**, and **SITE_CHAR**
 - Can copy and paste output from AERSURFACE
 - Can spatially vary by sector (up to 12)
 - Can Temporally vary by season/month
 - Need two sets data if both on-site and NWS data used

Stage 3 pathway: METPREP

- METPREP pathway
 - Can specify spatially varying surface characteristics
 - More detail later in AERSURFACE
 - FREQ_SECT: spatial & temporal changing patterns
 - Format: FREQ_SECT *frequency* *#sectors*
 - *Frequency* can be: ANNUAL, SEASONAL, or MONTHLY
 - *#sectors*: how many zones revolving around the center (12 max)
 - SECTOR: define spatial zones revolving around center
 - Format: SECTOR *sector_index* *beginning_direction* *ending_direction*
 - SITE_CHAR: provide surface characteristics data
 - Format: SITE_CHAR *freq_idx* *sector_idx* *albedo* *Bowen* *roughness*
 - *freq_idx*: # frequency in FREQ_SECT
 - *sector_idx*: # sector in SECTOR

Stage 3 pathway: METPREP

- METPREP pathway
 - **NWS_HGT**: specify height of surface instrument
 - Format: NWS_HGT WIND *height*
 - Height is in meters
 - METHOD: specify detailed processing method
 - Format: METHOD *variable option*
 - Several combinations available
 - METHOD WIND_DIR NORAND
 - Do not modify current wind direction
 - METHOD WIND_DIR RANDOM
 - Randomly +/- a few degrees to wind direction
 - NWS reports wind direction in nearest 10 degrees

Stage 3 pathway: METPREP

- METPREP pathway
 - METHOD (continue)
 - METHOD REFLEVEL SUBNWS
 - When on-site data used, but missing for certain hours, substitute missing on-site data with NWS data
 - When only NWS data used, this needs to be in inputs
 - METHOD STABLEBL ADJ_U*
 - Adjust U^* at low wind speed (sounds familiar?)
 - METHOD also has several other options, but we will skip them here. Check section 4.7.6 of AERMET user guide

Stage 2 example input

JOB

MESSAGES TestRun_3.MSG

REPORT TestRun_3.RPT

SURFACE

QAOUT MCO_2022_QAout.dat

UPPERAIR

QAOUT UA_2022_QAout_FSL_knot.dat

METPREP

XDATES 2022/1/1 2022/12/31

METHOD REFLEVEL SUBNWS

METHOD WIND_DIR RANDOM

NWS_HGT WIND 10

OUTPUT ORLANDO.SFC

PROFILE ORLANDO.PFL

FREQ_SECT ANNUAL 1

SECTOR 1 0 360

SITE_CHAR 1 1 0.2 1 0.5

Final remarks

- AERMET reads only “aermet.inp”
- For each stage, copy the input and rename to “aermet.inp”
 - Try use a “batch” file to do this automatically
- Do check the output data along the way
 - Max, min, diurnal variations